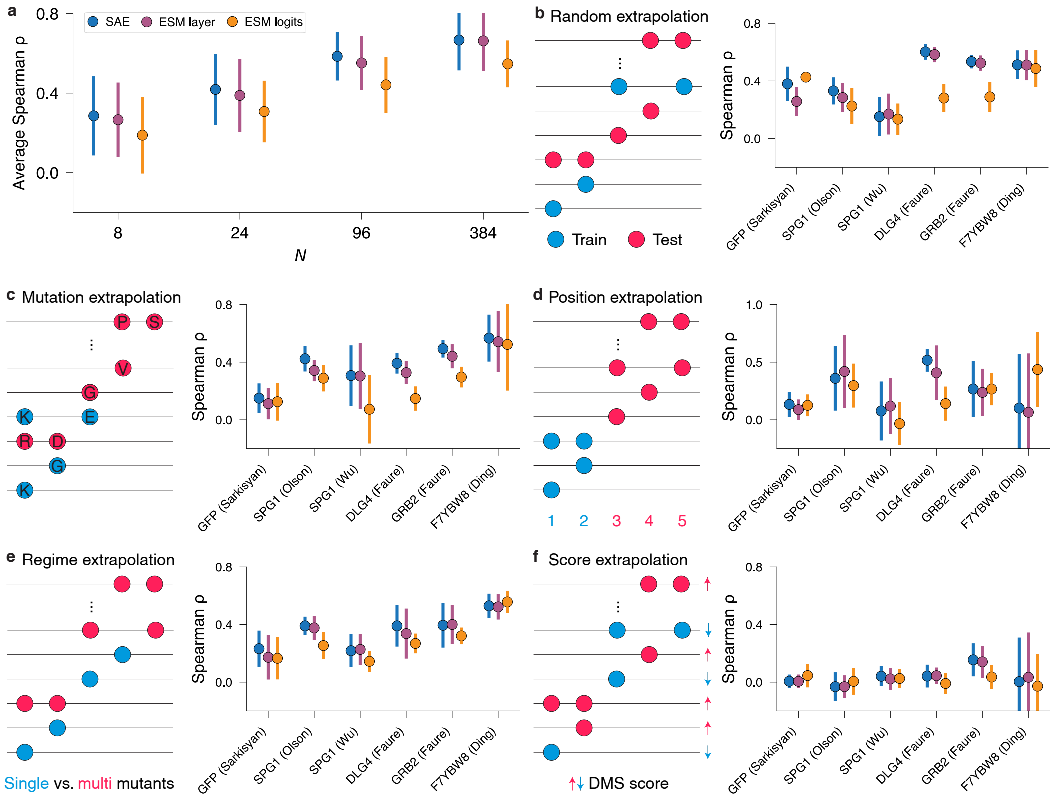

Figure 2: **Comparative performance of low-N fitness extrapolation in SAEs versus ESM2. a**, Average correlations of SAE, ESM layer, and ESM logits across all low-N regimes on random extrapolation. Fitness extrapolation correlations over each DMS assay using $N = 24$ sequences across **b**, random, **c**, mutation, **d**, position, **e**, regime, and **f**, score extrapolations. Error bars represent the standard deviation across nine independent runs with different random seeds.

3. **Position extrapolation**: We randomly designated 80% of amino acid positions as training positions (Fig. 2d). We then sampled N sequences for training that had mutations exclusively at the training positions. The other 20% of positions were held out as a test set.

4. **Regime extrapolation**: For DMS assays containing only single and double mutations, we trained on N single mutations and tested on all double mutations. For DMS assays with more than two mutations, we trained on N sequences drawn from single and double mutations, and tested on all sequences with more than two mutations (Fig. 2e).

5. **Score extrapolation**: We trained on N sequences with a fitness score lower than the wildtype and tested on all sequences with a fitness score higher than the wildtype (Fig. 2f).

Linear Probes. For each of these extrapolations, we trained a linear probe with Ridge regression on top of the SAE latent space. To benchmark against the performance of ESM2 without help from an SAE, we also trained linear probes on the ESM2 layer 24 embedding and ESM2 logits. For brevity, we refer to these methods as 1) SAE, 2) ESM layer, and 3) ESM logits. Following [14], we mean-pool the input to the linear probe over the respective embedding dimension. Formally, in SAEs, we denote $\bar{\mathbf{z}} \in \mathbb{R}^{d_{\text{SAE}}}$ to be the meaned activations of the latent space and $\mathbf{w} \in \mathbb{R}^{d_{\text{SAE}}}$ to be weights of the linear probe. The linear probe then computes the fitness score y via: $y = \mathbf{w}^T \bar{\mathbf{z}}$ (Fig. 1b). For all tasks, we set aside a portion of the training sequences to be used as validation. Further details are provided in Appendix A.3.

To ensure the robustness of our results, we ran a total of nine trials for each extrapolation. For random, position, and mutation extrapolations, we used three different random seeds to create the test set. For each of these test sets, we then randomly sampled N training sequences three times. For the regime and score extrapolations, where the test set is deterministic, we randomly sampled N training sequences nine times.

3.3 SAEs Achieve Improved Generalization to Unseen Variants Compared to ESM2

Fig. 2a shows the average Spearman correlation of SAE, ESM layer, and ESM logits under random extrapolation across all low-N regimes, while Table 2 breaks down results by DMS assay. SAEs achieve higher correlations than their ESM2 counterparts in 67% of random extrapolation experiments, and across all low-N regimes and fitness extrapolation tasks (Appendix B), they outperform in 58% of cases. These results suggest that SAE latents capture more biologically meaningful patterns and enable more reliable generalization to unseen variants. Among the different extrapolation settings, position, regime, and score extrapolation emerge as the most challenging, since they require the model to capture structural context and nonlinear interactions underlying protein function. Notably, SAEs outperform their ESM2 counterparts in 69% of position and regime extrapolation tasks, suggesting that their sparse latent space encodes fundamental biological constraints. We additionally notice that SAEs are not able to generalize to unseen variants when ESM2 does a poor job, such as in score extrapolation. This is not surprising, as intuitively, SAEs are trained to reorganize the information encoded by ESM2 into a more compact and disentangled representation. Thus, when the underlying pLM provides a limited predictive signal, the bottleneck in performance lies in the pLM rather than in the SAE.

Table 2: Average Spearman ρ across all low-N regimes under random extrapolation across each DMS assay. A full summary of results for other fitness extrapolation tasks is located in Appendix B.

Method	**DMS**	$N=8\uparrow$	$N=24\uparrow$	$N=96\uparrow$	$N=384\uparrow$
SAE	GFP_AEQVI_Sarkisyan	0.26 ± 0.15	0.38 ± 0.12	$\mathbf{0.56 \pm 0.05}$	$\mathbf{0.67 \pm 0.01}$
	SPG1_STRSG_Olson	0.16 ± 0.17	$\mathbf{0.33 \pm 0.09}$	$\mathbf{0.67 \pm 0.03}$	$\mathbf{0.82 \pm 0.01}$
	SPG1_STRSG_Wu	0.12 ± 0.16	0.15 ± 0.14	$\mathbf{0.34 \pm 0.04}$	0.35 ± 0.03
	DLG4_HUMAN_Faure	$\mathbf{0.45 \pm 0.16}$	$\mathbf{0.60 \pm 0.05}$	$\mathbf{0.67 \pm 0.03}$	$\mathbf{0.76 \pm 0.02}$
	GRB2_HUMAN_Faure	$\mathbf{0.31 \pm 0.15}$	$\mathbf{0.53 \pm 0.05}$	$\mathbf{0.63 \pm 0.02}$	$\mathbf{0.73 \pm 0.01}$
	F7YBW8_MESOW_Ding	$\mathbf{0.42 \pm 0.19}$	$\mathbf{0.51 \pm 0.10}$	0.64 ± 0.02	0.68 ± 0.02
ESM layer	GFP_AEQVI_Sarkisyan	0.21 ± 0.13	0.26 ± 0.10	0.46 ± 0.07	0.61 ± 0.01
	SPG1_STRSG_Olson	$\mathbf{0.17 \pm 0.22}$	0.28 ± 0.10	0.64 ± 0.02	0.81 ± 0.01
	SPG1_STRSG_Wu	$\mathbf{0.13 \pm 0.15}$	$\mathbf{0.17 \pm 0.14}$	0.31 ± 0.05	$\mathbf{0.36 \pm 0.07}$
	DLG4_HUMAN_Faure	0.40 ± 0.16	0.58 ± 0.05	0.66 ± 0.05	$\mathbf{0.76 \pm 0.02}$
	GRB2_HUMAN_Faure	0.28 ± 0.14	0.52 ± 0.05	0.60 ± 0.03	0.73 ± 0.02
	F7YBW8_MESOW_Ding	0.40 ± 0.15	0.51 ± 0.11	$\mathbf{0.65 \pm 0.02}$	$\mathbf{0.70 \pm 0.01}$
ESM logits	GFP_AEQVI_Sarkisyan	$\mathbf{0.31 \pm 0.12}$	$\mathbf{0.43 \pm 0.03}$	0.49 ± 0.05	0.57 ± 0.03
	SPG1_STRSG_Olson	0.12 ± 0.13	0.23 ± 0.13	0.42 ± 0.02	0.55 ± 0.01
	SPG1_STRSG_Wu	0.01 ± 0.13	0.14 ± 0.11	0.21 ± 0.08	0.30 ± 0.04
	DLG4_HUMAN_Faure	0.17 ± 0.17	0.28 ± 0.10	0.47 ± 0.05	0.60 ± 0.04
	GRB2_HUMAN_Faure	0.14 ± 0.10	0.29 ± 0.10	0.41 ± 0.07	0.59 ± 0.02
	F7YBW8_MESOW_Ding	0.38 ± 0.25	0.49 ± 0.13	0.65 ± 0.03	0.67 ± 0.02

Fig. 2b-f further illustrates the performance of SAE, ESM layer, and ESM logits across all extrapolation tasks with $N = 24$ sequences. We designated this as the smallest low-N regime for reliable extrapolation, with the SAE achieving an average correlation of 0.42. Across nearly all tasks, SAEs either match or outperform both ESM layers and ESM logits, highlighting their robustness and effectiveness in diverse low-N extrapolation settings. For additional results, see Appendix B.

4 SAEs for Low-N Protein Engineering

After demonstrating that SAEs are able to generalize to unseen variants, we then looked to assess their performance in generating high-functioning proteins (Fig. 1c). To explicitly optimize for function, we implemented a modified version of feature steering [28], which leverages the predictive scores from the linear probes. For all experiments, we used the linear probes trained on $N = 24$ sequences.

Table 3: Protein engineering results using $N = 24$ training sequences. All variants were constrained to a maximum of five mutations away from the wild type.

Method	DMS	Mean fitness ↑	Max fitness ↑	Top 10% fitness ↑	Top 20% fitness ↑
SAE	GFP_AEQVI_Sarkisyan	**3.49** ± 0.44	**3.87**	**3.75** ± 0.08	3.71 ± 0.07
	SPG1_STRSG_Olson	**2.75** ± 1.29	**4.53**	**4.47** ± 0.04	**4.29** ± 0.24
	SPG1_STRSG_Wu	**0.67** ± 0.94	**3.89**	**2.70** ± 0.79	**2.18** ± 0.76
	DLG4_HUMAN_Faure	**0.39** ± 0.22	**0.68**	**0.66** ± 0.02	**0.62** ± 0.05
	GRB2_HUMAN_Faure	**-0.10** ± 0.48	**0.67**	**0.59** ± 0.07	**0.49** ± 0.12
	F7YBW8_MESOW_Ding	0.81 ± 0.33	1.16	**1.15** ± 0.01	**1.13** ± 0.03
ESM layer	GFP_AEQVI_Sarkisyan	3.29 ± 0.66	3.72	3.71 ± 0.01	3.70 ± 0.01
	SPG1_STRSG_Olson	0.29 ± 1.95	3.19	2.74 ± 0.35	2.44 ± 0.39
	SPG1_STRSG_Wu	0.08 ± 0.30	1.69	0.81 ± 0.63	0.41 ± 0.60
	DLG4_HUMAN_Faure	-0.10 ± 0.41	0.63	0.45 ± 0.14	0.36 ± 0.13
	GRB2_HUMAN_Faure	-0.40 ± 0.39	0.30	0.24 ± 0.05	0.17 ± 0.10
	F7YBW8_MESOW_Ding	**1.06** ± 0.10	**1.16**	1.15 ± 0.02	1.12 ± 0.03
ESM logits	GFP_AEQVI_Sarkisyan	3.13 ± 0.86	3.76	3.73 ± 0.02	**3.72** ± 0.02
	SPG1_STRSG_Olson	-1.11 ± 2.21	2.27	2.05 ± 0.42	1.56 ± 0.60
	SPG1_STRSG_Wu	0.15 ± 0.37	1.69	1.13 ± 0.33	0.76 ± 0.50
	DLG4_HUMAN_Faure	-0.15 ± 0.38	0.53	0.36 ± 0.14	0.27 ± 0.13
	GRB2_HUMAN_Faure	-0.26 ± 0.44	0.58	0.44 ± 0.09	0.32 ± 0.14
	F7YBW8_MESOW_Ding	1.05 ± 0.06	1.12	1.11 ± 0.01	1.11 ± 0.01
Random	GFP_AEQVI_Sarkisyan	3.36 ± 0.70	3.75	3.72 ± 0.02	3.70 ± 0.02
	SPG1_STRSG_Olson	-1.25 ± 2.63	3.05	2.40 ± 0.49	1.99 ± 0.55
	SPG1_STRSG_Wu	0.33 ± 0.76	3.61	2.27 ± 0.82	1.36 ± 1.11
	DLG4_HUMAN_Faure	-0.34 ± 0.43	0.35	0.32 ± 0.04	0.24 ± 0.10
	GRB2_HUMAN_Faure	-0.96 ± 0.38	0.16	-0.13 ± 0.18	-0.30 ± 0.22
	F7YBW8_MESOW_Ding	0.66 ± 0.37	1.16	1.11 ± 0.03	1.06 ± 0.06

4.1 Experimental Setup

For feature steering, we first identified the most predictive latent features by examining the largest-magnitude weights of the linear probe. For each of these high-impact latents, we increased its activation by a hyperparameter multiplier. The updated latent representation was then passed through the SAE decoder and fed into ESM2 to design a new sequence. We optimized the multiplier by selecting the value that yielded the highest predicted fitness score from the linear probe. Similar to fitness extrapolation, to benchmark against the performance of ESM2, we also designed sequences using the linear probes trained on the ESM layer and ESM logits via simulated annealing, following the procedure detailed in [27]. Additionally, we included a random baseline by generating sequences with a random number of mutations and amino acid substitutions. Further details on our experimental setup are provided in Appendix A.4.

We used a multi-layer perceptron (MLP) trained on the DMS assays to evaluate the fitness of our designed variants (see Appendix A.5). To constrain our search space and ensure the MLP's predictions are a good proxy for experimental fitness, we limited all designed variants to a maximum of five mutations away from the wildtype. A notable exception to this setup is the SPG1_STRSG_Wu DMS: this assay provides ground-truth fitness values for all possible combinatorial variants over four positions. Therefore, we directly used the fitness values from SPG1_STRSG_Wu to evaluate our designed variants and limited our maximum number of mutations to four. A total of 50 variants were designed per DMS assay.

4.2 SAEs Design High-functional Variants and Capture Biological Motifs

Table 3 shows the performance of all methods in generating highly-functional variants. Across all metrics and DMS assays, SAEs outperform their ESM2 counterparts in 88% of cases. More specifically, our SAE steering approach designed the top fitness variants in five out of the six DMS assays. Additionally, SAE steering designed the highest top 10% fitness variants across all DMS assays and the highest top 20% variants in five out of six DMS assays. This suggests that SAE steering is not only capable of discovering the single top-performing variant, but also is capable of generating a diverse pool of highly functional variants.